



THARAN S M

CONTACT

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EDUCATION

SRM Valliammai Engineering College

Bachelor of Engineering ECE
2023 - 2027

Shree Niketan Matriculation Hr. Sec. School

HSE
2021 - 2023

SKILLS

Languages

- Verilog
- Python
- Shell/Bash Scripting
- C/C++

Software & Tools

- Xilinx Vivado
- OrCAD PSpice
- OrCAD Capture CIS
- Cadence

Methodology

- Low-Power Design
- RTL Verification

LANGUAGE

- Tamil
- English

EXECUTIVE SUMMARY

To secure a Hardware Project Engineer Trainee role, leveraging a strong background in analog/digital electronics, low-power digital systems, and FPGA based design. Eager to apply knowledge of circuit schematics, PCB design concepts, and hardware debugging to contribute to impactful electronic projects while expanding technical depth from digital arithmetic hardware to complete embedded systems.

PROJECTS

16-Bit Multi-Cycle Processor with Adaptive ALU and Booth MAC

Jan2026 - May2026

Led the core logic development and RTL design of a power-optimized 16-bit RISC processor architecture. The project focused on engineering an adaptive ALU and a Booth multiplier/MAC unit to significantly reduce dynamic power consumption while improving overall processing speed. By managing both the hardware implementation and rigorous verification processes (synthesized in Vivado), The project successfully delivered a highly efficient, value driven computational solution.

Smart Blind Stick – Smart India Hackathon (SIH) 2025

Sep2025 - Dec2025

- Directed the hardware-software integration of a GPS-enabled assistive device as the project team lead.
- Interfaced embedded sensors, including ultrasonic obstacle detection, to develop a robust safety tool for the visually impaired.
- Engineered a reliable communication pipeline between the embedded hardware and a guardian mobile application to trigger real-time SOS alerts.

FSM-Based Digital Controller Design

Oct2024 - Dec2024

Designed and verified FSM based digital controllers using Verilog HDL and RTL design methodologies. Implemented both Moore and Mealy architectures to optimize control logic, and ensured system reliability by debugging state transitions through rigorous testbench simulations in Xilinx Vivado.

CERTIFICATIONS

FPGA Design Flow & TCL Scripting in Vivado | SenseAcademia [Mar 2026]

Focused on automating hardware design workflows and mastering FPGA architecture skills directly applicable to firmware automation and hardware testing.

Industry Oriented FIFO Design in VLSI & FPGA | SenseAcademia [Mar 2026]

Developed high-speed data buffering logic, essential for communication protocols (I2C/SPI/UART) and hardware-software synchronization.